

DİCLE ÇOBAN HW8 REPORT

No: 220104004088

Variables:

people: A map storing people in the network with their names as keys and Person objects as values.

friendships: A map storing friendships between people, where each person is mapped to a list of their friends.

Methods:

addPerson(String name, int age, List<String> hobbies): Adds a new person to the network with the given attributes.

removePerson(String name): Removes a person from the network.

addFriendship(String name1, String name2): Establishes a friendship between two people.

removeFriendship(String name1, String name2): Removes a friendship between two people.

findShortestPath(String startName, String endName): Finds the shortest path between two people using BFS.

countClusters(): Counts and prints the clusters of connected people in the network.

suggestFriend(String name, int maxSuggestions): Suggests friends for a person based on mutual friends and common hobbies.

Helper Methods:

calculateScore(Person person, Person friend): Calculates the friend suggestion score.

countMutualFriends(Person person1, Person person2): Counts the number of mutual friends between two people.

countCommonHobbies(Person person1, Person person2): Counts the number of common hobbies between two people.

bfs(Person start, Set<Person> visited, List<Person> cluster): Performs BFS to find all people in a cluster.

PS: I didn't add timestamp while prompting a user because the name of the object already acts as a key in the map so its automatically created uniquely so the user can't create another object with the same name.

Also in the example main class there isn't any timestamp for adding a friendship.

Adding a Person:

----- Social Network Menu -----

1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit

Enter your choice: 1

Enter name: *Emily Davis*

Enter age: 28

Enter hobbies (comma-separated): *running,swimming*

Person added: Emily Davis (Age: 28, Hobbies: [running, swimming]) (Wed May 29 20:51:10 TRT 2024)

----- Social Network Menu -----

1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit

Enter your choice: 1

Enter name: *Frank Wilson*

Enter age: 26

Enter hobbies (comma-separated): *reading,hiking*

Person added: Frank Wilson (Age: 26, Hobbies: [reading, hiking]) (Wed May 29 20:51:23 TRT 2024)

----- Social Network Menu -----

1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit

Enter your choice: 1

Enter name: *Alice Johnson*

Enter age: 27

Enter hobbies (comma-separated): *hiking,painting*

Person added: Alice Johnson (Age: 27, Hobbies: [hiking, painting]) (Wed May 29 20:50:40 TRT 2024)

----- Social Network Menu -----

1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit

Enter your choice: 1

Enter name: *Bob Brown*

Enter age: 30

Enter hobbies (comma-separated): *reading,swimming*

Person added: Bob Brown (Age: 30, Hobbies: [reading, swimming]) (Wed May 29 20:50:55 TRT 2024)

```

----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 1
Enter name: John Doe
Enter age: 25
Enter hobbies (comma-separated): reading,hiking,cooking
Person added: John Doe (Age: 25, Hobbies: [reading, hiking, cooking]) (Wed May 29 20:50:07 TRT 2024)

```

→ Also adding a person

```

----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 1
Enter name: Jane Smith
Enter age: 22
Enter hobbies (comma-separated): swimming,cooking
Person added: Jane Smith (Age: 22, Hobbies: [swimming, cooking]) (Wed May 29 20:50:24 TRT 2024)

```

I add 6 people same people in the example main class

Adding Friendship:

```

----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 3
Enter name of first person: John Doe
Enter name of second person: Jane Smith
Friendship added between John Doe and Jane Smith

```

```

----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 3
Enter name of first person: John Doe
Enter name of second person: Alice Johnson
Friendship added between John Doe and Alice Johnson

```

John Doe → Jane Smith → Bob Brown
 John Doe → Alice Johnson

```

----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 3
Enter name of first person: Bob Brown
Enter name of second person: Jane Smith
Friendship added between Bob Brown and Jane Smith

```

```

----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 3
Enter name of first person: Emily Davis
Enter name of second person: Frank Wilson
Friendship added between Emily Davis and Frank Wilson

```

Emily Davis → Frank Wilson

Showing the Cluster:

I already wrote the cluster below the adding friendship photos in the previous page.

```
----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 7
Cluster 1: [Emily Davis (Age: 28, Hobbies: [running, swimming]), Frank Wilson (Age: 26, Hobbies: [reading, hiking])]
Cluster 2: [Alice Johnson (Age: 27, Hobbies: [hiking, painting]), John Doe (Age: 25, Hobbies: [reading, hiking, cooking]), Jane Smith (Age: 22, Hobbies: [swimming, cooking]), Bob Brown (Age: 30, Hobbies: [reading, swimming])]
Total clusters: 2
```

Suggesting friends :

```
----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 6
Enter name of person: John Doe
Enter maximum number of suggestions: 5
Top 5 suggested friends for John Doe:
- Bob Brown (Score: 1.5, Mutual Friends: 1, Common Hobbies: 1)
- Frank Wilson (Score: 1.0, Mutual Friends: 0, Common Hobbies: 2)
- Alice Johnson (Score: 0.5, Mutual Friends: 0, Common Hobbies: 1)
- Jane Smith (Score: 0.5, Mutual Friends: 0, Common Hobbies: 1)
- Emily Davis (Score: 0.0, Mutual Friends: 0, Common Hobbies: 0)
```

Finding the shortest path:

```
----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 5
Enter name of start person: John Doe
Enter name of end person: Bob Brown
Shortest path: [John Doe (Age: 25, Hobbies: [reading, hiking, cooking]), Jane Smith (Age: 22, Hobbies: [swimming, cooking]), Bob Brown (Age: 30, Hobbies: [reading, swimming])]
```

John Doe and Bob Brown have a mutual friend

so John Doe → Jane Smith → Bob Brown

Remove a Person:

```
----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 2
Enter name of person to remove: Bob Brown
Person removed: Bob Brown (Age: 30, Hobbies: [reading, swimming])
```

```
----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 3
Enter name of first person: Bob Brown
Enter name of second person: John Doe
One or both persons not found in the network.
```

→ I delete Bob Brown
thats why I couldn't add
friendship.

Remove Friendship:

```
----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 4
Enter name of first person: Emily Davis
Enter name of second person: Frank Wilson
Friendship removed between Emily Davis and Frank Wilson
```

Showing the cluster :

to show that I removed a Friendship.

```
----- Social Network Menu -----
1. Add person
2. Remove person
3. Add Friendship
4. Remove Friendship
5. Find shortest path
6. Suggest friend
7. Count clusters
8. Exit
Enter your choice: 7
Cluster 1: [Emily Davis (Age: 28, Hobbies: [running, swimming])]
Cluster 2: [Alice Johnson (Age: 27, Hobbies: [hiking, painting]), John Doe (Age: 25, Hobbies: [reading, hiking, cooking]), Jane Smith (Age: 22, Hobbies: [swimming, cooking])]
Cluster 3: [Frank Wilson (Age: 26, Hobbies: [reading, hiking])]
Total clusters: 3
```

There are three cluster now because before removing the friendship Frank and Emily was friends, now both of them don't have friends they are clusters separately.

That's why cluster number increased.